

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

21st March 2017

Subject CT1 – Financial Mathematics

Time allowed: Three Hours (10.30 – 13.30 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. *Mark allocations are shown in brackets.*
3. *Attempt all questions, beginning your answer to each question on a separate sheet.*
4. *Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately.

- Q. 1) i)** Describe the difference between following types of asset:
- a)** Debenture stocks and
 - b)** Unsecured loan stocks. (2)
- ii)** List reasons why investing in property may not be preferred alternative to investing in Government securities. (2)
- iii) a)** How does Government issue a loan by tender? (2)
- b)** What are the main advantages and disadvantages to the Government and to investors of issuing bonds at a set price as opposed to by tender? (2)
 - c)** Give the name of three securities, which have uncertain income and give reason for such uncertainty. (3)
- [11]**
- Q. 2) i)** The rate of discount per annum convertible quarterly is 6%. Calculate:
- a)** The equivalent rate of interest per annum convertible half yearly. (2)
 - b)** The equivalent rate of discount per annum convertible monthly. (2)
- ii)** On 15th April, 2005 Amit borrowed Rs 2,00,000 to be repaid one year later by single payment of Rs 2,20,000.
- Amit repaid the loan early on 17th July, 2005.
- a)** Find the sum paid by Amit to terminate the contract assuming that the interest is reduced proportionately for early settlement. (1.5)
 - b)** Calculate APR on the completed transaction (1.5)
- [7]**
- Q. 3)** A Government owned company, ONGC has just purchased an oil rig for Rs 2.7 crores along with the drilling rights for three years in a section of Arabian Sea. The rig will be operated for three years at an annual cost of Rs 20,00,000 payable at the end of each year. At the end of third year the rig will be sold for Rs 10,00,000.
- At most one oil strike can occur in any year and the probabilities of striking oil in first, second and third years are 0.25, 0.2 and 0.1 respectively. These probabilities are independent of each other. The company assumes that there will be a continuous output from each strike at the rate of 50,000 barrels per year for ten years commencing at the end of the year of the strike.
- i)** Assuming a constant oil price of Rs 200 per barrel, find the effective annual rate of interest at which the present value of the company's outlays equals the expected present value of its income from the project. (5)

- ii) If successful oil strike occurs only in the third year and oil price following that remains constant at Rs 170 per barrel, calculate the Net present value (NPV) of the project at 10% p.a. effective rate of return.
Also comment whether it is profitable to invest in the project.

(3)
[8]

- Q. 4)** The ABC trust runs the investment fund which had following fund value at the various dates:

Date	Fund Value (in Rs crores)
1 st January, 2011	30.50
30 th September, 2011	33.00
31 st December, 2011	38.50
1 st July, 2012	41.05
31 st December, 2012	45.00
30 th June, 2013	45.60
31 st December, 2013	47.00

Over the three years period the investment fund has the following inflows and outflows:

Date	In-flows (in Rs crores)	Out-flows (in Rs crores)
1 st October, 2011	6.00	-
30 th June, 2012	4.50	-
1 st July, 2013	-	2.50

- i) Calculate the annual effective time weighted rate of return (TWRR) and money weighted rate of return (MWRR) earned on the fund from 1st January, 2011 to 31st December, 2013.
- ii) Calculate the Linked rate of return from 1st January, 2011 to 31st December, 2013.
- iii) Why TWRR is better measure of investment performance than MWRR.

(5)
(5)
(1)
[11]

- Q. 5)** The n -year spot rate of interest, y_n , is given by:

$$y_n = 0.04 + \frac{n}{1000} \text{ for } n = 1, 2 \text{ and } 3$$

- i) Calculate the implied one-year forward rates applicable at times $t = 1$ and $t = 2$.
- ii) Assuming no arbitrage applies, calculate the price at time $t = 0$ per Rs 1,000 nominal of a bond which pays annual coupons of 3% in arrears and is redeemed at 110% after 3 years.

(4)
(3)
[7]

- Q. 6)** i) While valuing future payments an investor uses the formulae

$$v(t) = \alpha (\alpha + 1) / ((\alpha + t)(\alpha + t + 1)) \quad t \geq 0$$
where, $v(t)$ is the discounting factor at time t and α is a positive constant

Show that the force of interest (δ) at time t will be

$$\delta(t) = (2t + 2\alpha + 1) / ((\alpha + t)(\alpha + t + 1))$$

(3)

- ii) For a bank deposit over a given year, the force of interest per annum was 15% at the start of year, 10% at middle of the year, and 8% at the end of year. The force of interest per annum follows a quadratic function given by
- $$\delta(t) = a + bt + ct^2$$
- where, a, b, c are constants and t is the time in years.

Find the accumulated amount at the end of year of a deposit of Rs 1,50,000 at the start of year.

(5)
[8]

- Q. 7) i) An annuity payable annually in arrears for n years is such that the payment at time t is $(n+1)t - t^2$. Show that the present value of the annuity may be expressed as
- $$(n+1)(Ia)_n - \frac{2(Ia)_n - a_n - n^2 v^{n+1}}{1-v}$$

(4)

- ii) An investor is entitled for an annuity of Rs 10,000 per annum payable annually in arrears for 10 years. The investor is under an obligation to make two payments of Rs 30,000 in five years' time and 10 years' time.

a) Using the effective rate of interest 10% p.a. find the present value and discounted mean term of the payment received and payment to be made by the investor.

(4)

b) Hence find the discounted mean term of the investor's net cashflow using the effective rate of interest 10% p.a.

(1)
[9]

- Q. 8) An insurance company is holding a portfolio of securities to back it with profit fund. The effective annual returns (i_t) are independent and $(1 + i_t)$ is log-normally distributed, where i_t is the return in the t^{th} year. The mean and standard deviation of the effective annual return (i_t) are 6% and 8% respectively.

i) Calculate the expected value of an investment of Rs 1 crore after 10 years

(2)

ii) Calculate the probability that the accumulation of the investment will be less than 90% of the expected value.

(6)
[8]

- Q. 9) Exactly ten years ago a 25 year loan was issued with effective rate of interest 8% p.a. payable annually in arrear. The loan had to be repaid with a level installment amount of Rs 1000 per annum.

So far all the ten due payments have been made. Now an investor is considering the purchase of the remaining installments from the loan issuer. The investor is liable to pay income tax at the rate of 40% on the interest content (according to the original loan schedule) of each payment. What price should the investor pay to obtain a net yield of 10% per annum effective?

[7]

Q. 10) i) An insurance company has liabilities to meet which involve a series of cash payment evenly spread over next ten years. All its investments are currently held in cash. Describe the form of swap contract which could help the insurance company immunise its liabilities. (2)

ii) A three year forward contract is to be issued on a particular company share. The current market value of the share is Rs 450 and a dividend of Rs 20 per share has just been paid.

The quarterly dividends are expected to increase by 1% per quarter for the first two years and by 1.5% per quarter for the third year.

Assuming a risk-free force of interest of 5% per annum and no arbitrage opportunities, calculate the forward price of the contract. (7)

[9]

Q. 11) Mr. Sam wishes to purchase a house and has decided to take the loan from a bank to finance the purchase. The bank agrees to finance the purchase with the following terms:

- 80% of the house cost will be financed by the bank and the remaining cost will be borne by Mr. Sam.
- The loan is repayable by an increasing annuity, payable annually in arrears for 15 years. The repayment at the end of first year is Rs. 2,00,000 and subsequent repayments increase by Rs. 20,000 every year.

The repayments were calculated using a rate of interest of 12% per annum effective.

i) Calculate the amount of loan granted by the bank and the total cost of the house. (4)

ii) Construct the schedule of repayment for years nine and ten (i.e. immediately after the 8th repayment installment), showing the outstanding loan at the beginning of the year, the repayment installment, interest portion and capital repayment portion in each year. (5)

iii) Immediately after the tenth installment of repayment, Mr. Sam wishes to transfer its balance loan to another bank which offered him the loan at the effective rate of 10% per annum with regular annual payments of fixed amount. The new bank charges the upfront processing fees of 1% of loan outstanding. The other terms of the loan are unchanged.

Calculate the revised annual payments for Mr. Sam.

Based on your calculations comment whether it is advisable for Mr. Sam to transfer its loan to new bank. (6)

[15]
